10. (a) The enthalpy of neutralisation of an acid is defined as the enthalpy change when 1 mol of aqueous H^+ ions is neutralised by aqueous OH^- ions according to the equation shown. The reaction is exothermic.

$$H^+(aq) \quad + \quad OH^-(aq) \longrightarrow H_2O(l)$$

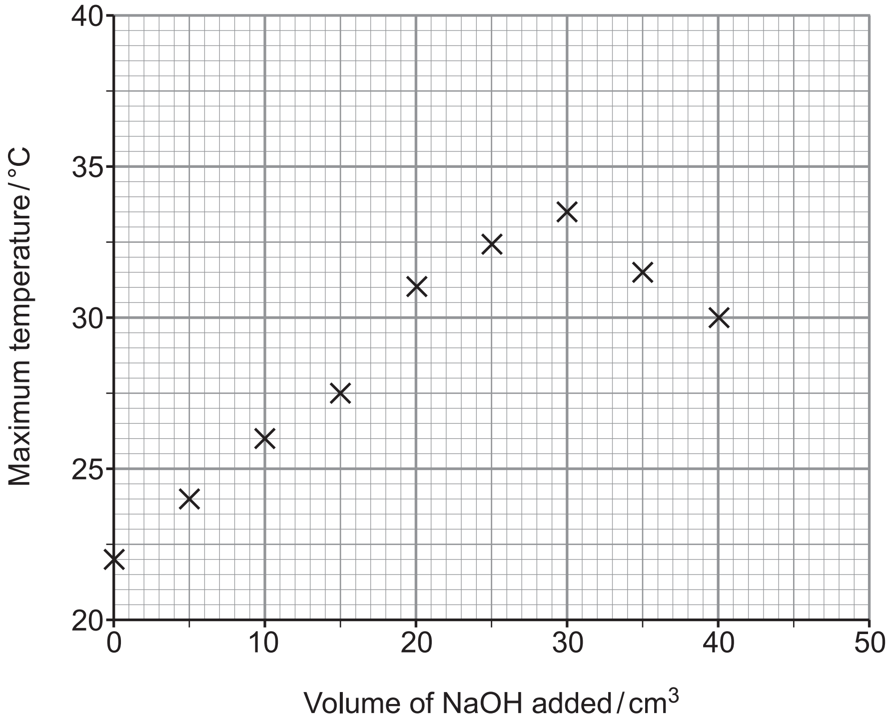
Some students followed the instructions below to determine the enthalpy change of neutralisation of methanoic acid, HCOOH.

1. Weigh 24.7 g of methanoic acid and mix with water to make 250 cm^3 of solution. Record the temperature of the solution.
2. Transfer eight 25.0 cm^3 portions of this solution into eight insulated cups.
3. Using a burette add 5.0 cm^3 of aqueous sodium hydroxide to the solution in the first cup. Stir and record the maximum temperature reached.
4. Add the following volumes of aqueous sodium hydroxide to each of the remaining cups in turn:

   10.0 cm^3 15.0 cm^3 20.0 cm^3 25.0 cm^3 30.0 cm^3 35.0 cm^3 40.0 cm^3

   Stir and record the maximum temperature reached in each cup.
5. Plot a graph of maximum temperature reached against volume of sodium hydroxide added.

Their results are plotted in the graph.



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(i) Name the apparatus used to transfer exactly 25.0 cm^3 of methanoic acid solution into the insulated cups. [1]

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(ii) State why a higher maximum temperature is recorded when increasing volumes of sodium hydroxide are added. [1]

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(iii) Explain why the maximum temperature recorded decreases when more than 30 cm^3 of sodium hydroxide is added. [2]

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(iv) On the graph, draw **one** straight line through the points that show an increase in maximum temperature and **another** straight line through the points that show a decrease in maximum temperature. [1]

(v) From the graph, deduce the volume of sodium hydroxide needed to neutralise 25.0 cm^3 of the methanoic acid solution and the temperature increase at that point.

Assume that the initial temperature of every 25.0 cm^3 of methanoic acid solution is 22.0 °C. [2]

Volume = ... cm^3

Temperature increase = ... °C

2410U201 09

Examiner only

(vi) Use your answers to part (v) to calculate the amount of heat released by the neutralisation reaction. [2]

Heat released = ... J

(vii) Calculate the number of moles of methanoic acid in 25.0 cm^3 of the solution and hence the enthalpy change of neutralisation of methanoic acid. [3]

Enthalpy change of neutralisation = ... $kJ\,mol^{-1}$

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(b) The experiment is repeated using hydrochloric acid instead of methanoic acid and a more negative value of the enthalpy change of neutralisation is calculated.

Suggest and explain a reason for this difference. [2]

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(c) (i) Write the equation for the reaction that occurs when solid copper(II) carbonate is added to aqueous methanoic acid to form aqueous copper(II) methanoate, $(HCOO)_2Cu$.

Include state symbols. [2]

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(ii) State what is observed during the reaction in part (i). [2]

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2410U201 11

18